SPRING **MID SEMESTER EXAMINATION-2025**

School of Computer Engineering
Kalinga Institute of Industrial Technology, Deemed to be University
ARM and Advanced Microprocessors
[EC-30007]

Time: 1 1/2 Hours **. Full Mark: 20**

Answer Any four questions including question No.1 which is compulsory.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. Answer all the questions. [1 Mark X 5]

   a) What is the maximum physical address space that the 80286 can access in Real Mode and Protected Mode?

   b) How many descriptors are stored in the GDT of the 80386 if the limit value stored in the GDTR register is 07FFH and the base address is 05300000H?

   c) What does the Granularity (G) bit of the 80386 specify?

   d) What are the improvements in the 80486 over the 80386?

   e) Calculate the average memory access time in an 80486 system, given that the DRAM access time is 80 ns, the cache access time is 10 ns, and the cache hit ratio is 85%.

2. Write short notes on Real Address mode and Protected Virtual Address mode (PVAM).

   Write an assembly language program to set the Protection Enable (PE) bit in the MSW register for switching from Real Mode to Protected Mode.

   What is need for memory management in high-level processors?
   Mention two limitations of the 80286 [2+1+2=5 Marks]

3. Discuss the translation scheme for converting a logical address to a physical address in the 80286 processor in detail, when TI = 0.

   In an 80386, if the limit and the base in the GDTR are 0FFFH and 00200000H, respectively, what is the beginning address of the table, the size of the table in bytes, and the ending address of the table? [4+1 Marks]

4. Give the format of the 8-byte 80286 Descriptor [3+2 Marks]

Identify the following, if a segment descriptor has an access right byte equal to F4H:

1. Is the memory segment present in physical memory?
2. Is it a data segment or a code segment?
3. What is the privilege level of the memory segment?
4. Has the memory segment been accessed previously by any other application program?

5. What is the TLB in the 80386? Discuss the paging mechanism with TLB. [4+1 Marks]

In an 80386 microprocessor, if the status of the byte enable pins during a data write bus cycle is $\overline{BE_3},\overline{BE_2},\overline{BE_1},\overline{BE_0}$ =1101, over which data lines will the byte data get transferred?.

*** Best of Luck ***